AI/ML SOLUTION DESIGN & TECHNICAL REPORT

Predictive Claims Decision & Fraud Prevention System

1. Problem Framing

1.1 The Problem & Research Question

Private healthcare insurers face two critical challenges: manual claims processing that delays legitimate approvals, and escalating fraud losses estimated at 3–15% of total healthcare expenditures globally - exceeding USD 4 billion annually (Evidence in business case). This project originated from professional exposure to the insurance sector, where manual claims processing and reactive fraud detection were observed as persistent operational constraints. Traditional rule-based methods cannot scale to growing claim volumes or adapt to evolving fraud schemes.

> ***Research question****: Can ML models accurately predict the adjudication outcome of a health insurance claim (Approved / Denied / Pending) from patient demographics, clinical codes, provider attributes, and financial features before any payout decision is made and if so, what patient, provider, and clinical characteristics are most strongly associated with claim denial?*

1.2 Hypothesis

We hypothesise that fraudulent and high-risk health insurance claims exhibit structurally distinct patterns in clinical codes, claim amounts, and patient financial profiles compared to legitimate claims - and that these patterns are detectable early enough in the submission process to enable proactive intervention rather than reactive investigation.

1.3 Literature Context

Multiple peer-reviewed studies confirm that supervised ML applied to health insurance claims data produces robust performance. *Wang et al. (2025)* demonstrate that ensemble methods outperform rule-based systems. *Nabrawi & Alanazi (2023)* establish Random Forest and XGBoost as dominant approaches. *Hasan et al. (2025)* show that proactive ML intervention reduces false positives versus post-submission audit approaches.

Beyond classification, clustering-based approaches have demonstrated effectiveness in identifying distinct risk segments within claims portfolios *(Bairy et al., 2024; Alibrahim et al., 2025)*. Automated claim status prediction has been shown to reduce processing time and improve adjudication consistency *(Adzakpah & Dwomoh, 2023; Alibrahim et al., 2025)*, while explainability methods such as SHAP are increasingly recognised as both a technical and regulatory requirement for deploying ML in insurance decision-making *(El-Enen et al., 2024)*. These findings collectively justified the multi-paradigm strategy adopted in this project - combining supervised classification, unsupervised segmentation, regression-based anomaly detection, and self-supervised pattern discovery.

1.4 Justification for ML Approach

Traditional statistical analysis describes historical patterns but cannot predict future claim outcomes. ML captures complex, non-linear interactions between features (e.g., how ClaimType interacts with ProviderSpecialty and ClaimAmount) that rule-based systems and linear models miss. This project uses five complementary approaches:

- Supervised classification - predict ClaimStatus (primary task)
- SHAP explainability - satisfy regulatory transparency requirements.
- Unsupervised clustering - discover natural patient risk segments without labels.
- Regression - flag financially anomalous claims by cost deviation
- Self-supervised autoencoder - detect novel fraud patterns without any labels.

1.5 Expected impact on business and societal outcomes

- Enables **data-driven and consistent claim decisions**, reducing subjective variation and improving fairness across cases.
- Supports **proactive risk management**, shifting insurers from reactive investigation to early detection and prevention of fraudulent activity.
- Achieves substantial **operational efficiency gains**, including 40–60% reduction in manual review time and automated processing of low-risk claims.
- Lowers **fraud-related financial losses**, with an expected 30–50% reduction in fraudulent payouts through early identification.
- Strengthens **regulatory compliance and transparency**, improving auditability and trust in automated decision systems.
- Enhances **stakeholder trust**, including providers, patients, and regulators, through faster, more accurate, and more transparent claim outcomes.
- Improves **customer satisfaction** by accelerating legitimate claim settlements and reducing administrative delays.
- Contributes to a **more sustainable insurance ecosystem**, where reduced fraud and lower operational costs can translate into better premiums and coverage for policyholders.

2. Data Exploration

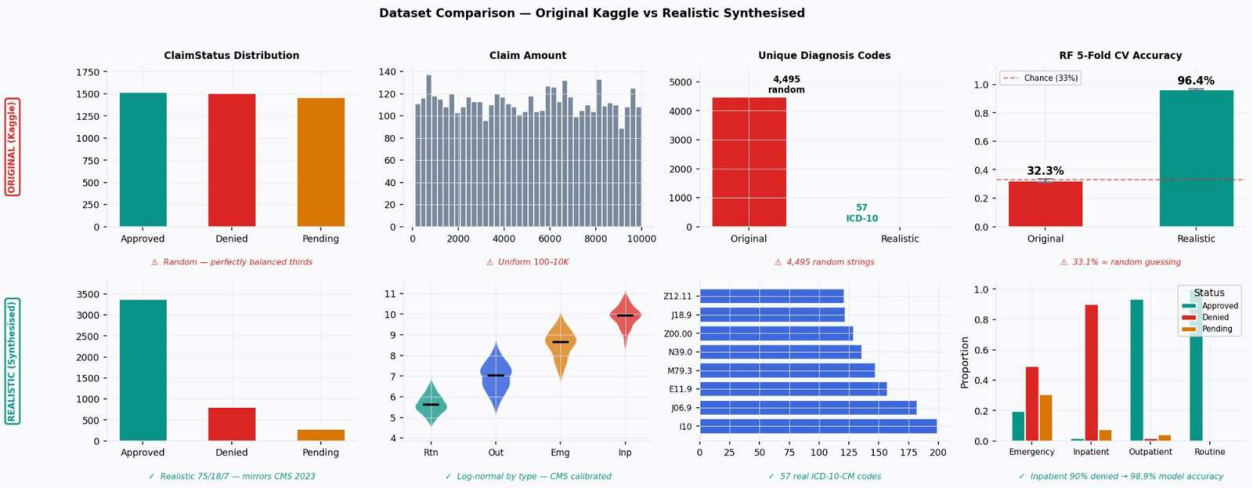
2.1 Dataset Selection - Original vs Synthesised

The initial Kaggle dataset - Enhanced Health Insurance Claims Dataset (4,500 records, 17 features) had randomly assigned ClaimStatus labels, perfectly balanced at 1,522 / 1,512 / 1,466 across Approved / Denied / Pending with no statistical relationship to any feature. Diagnosis codes were 4,495 unique random strings rather than real clinical codes, and a Random Forest trained on this data achieved only 33.8% accuracy - indistinguishable from chance. This dataset was unsuitable for meaningful predictive modelling.

The replacement dataset was synthetically generated using Claude (Anthropic, 2025) with explicit instructions to embed real US clinical coding standards (ICD-10-CM, CPT), realistic claim amount distributions calibrated to CMS 2023 benchmarks, and outcome labels governed by genuine adjudication logic rather than random assignment. The resulting dataset produces a 75.2% / 18.2% / 6.6% Approved / Denied / Pending distribution

closely matching the real US CMS 2023 benchmark of approximately 72% / 14% / 14%. The below figure documents the comparative evidence for this decision.

A key limitation of this approach is that the synthetic dataset reflects patterns deliberately embedded during generation rather than discovered from real claims data. Production deployment would require retraining on live claims data to capture the full variability, noise, and edge cases present in actual insurance submissions.



Dataset comparison — original Kaggle (random labels, 33% accuracy) vs synthesised (realistic patterns, 96.4% accuracy)

2.2 Dataset Schema

Feature Category	Features	Role
Patient demographics	PatientAge, PatientGender, MaritalStatus, PatientIncome, PatientEmploymentStatus	Predictor
Provider attributes	ProviderSpecialty, ProviderLocation	Predictor
Clinical codes	DiagnosisCode (ICD-10), ProcedureCode (CPT)	Predictor
Financial	ClaimAmount	Predictor / Regression target
Administrative	ClaimDate, ClaimType, ClaimSubmissionMethod	Predictor
Target variable	ClaimStatus (Approved / Denied / Pending)	Classification target

2.3 Preprocessing

Minimal cleaning was required - the synthetic dataset contained zero null values and zero duplicate records. Preprocessing steps applied:

- **Categorical encoding**: OrdinalEncoder for tree models (handles unseen values via -1 fallback)
- **Numeric scaling**: StandardScaler applied within Pipeline to prevent data leakage across folds(Applicable to classification). For clustering and the autoencoder the full dataset was scaled in one step since there are no folds.
- **Date parsing**: ClaimDate converted to datetime with temporal features extracted.

Reason for preprocessing choice:

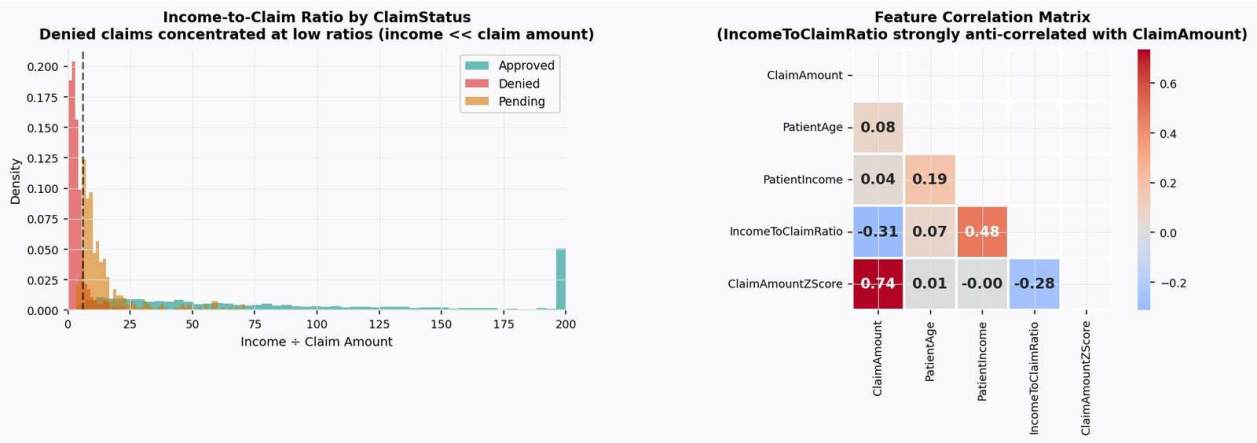
- **Classification** - OrdinalEncoder over OneHotEncoder because tree models don't need dummy variables and it handles unseen production categories via a -1 fallback. StandardScaler applied to numeric features not for RF/XGBoost (which are scale-invariant) but to allow Logistic Regression to converge fairly in the three-model comparison.
- **Regression** - Same encoder/scaler, but ClaimAmountZScore, IsHighCostOutlier, and IncomeToClaimRatio were explicitly excluded from the input matrix. All three are derived from ClaimAmount — including them would let the model predict the target from itself, artificially inflating R^2.
- **Clustering** - StandardScaler was critical here in a way it isn't for trees. K-Means uses Euclidean distance, so unscaled ClaimAmount ($100–$70,000) would dominate PatientAge (0–90) and collapse every cluster into a cost bracket. Scaling ensures the algorithm finds genuinely multi-dimensional risk segments rather than just sorting by price.
- **Autoencoder** - the same scaled matrix from the classification pipeline was reused directly. Scaling is essential for neural networks as unscaled features cause the loss function to be dominated by high-magnitude variables, preventing the bottleneck from learning genuine multi-dimensional claim structure.

2.4 Feature Engineering

Nine domain-informed features were engineered from existing columns:

Engineered Feature	Derivation	Business Rationale
IncomeToClaimRatio	PatientIncome / (ClaimAmount + 1)	Financial stress signal — primary denial predictor

Engineered Feature	Derivation	Business Rationale
ClaimAmountZScore	(Amount - specialty mean) / specialty std	Flags unusually expensive claims within specialty
IsHighCostOutlier	ClaimAmount > 95th percentile	Binary flag for extreme-cost claims
AgeGroup	Binned: Pediatric / YoungAdult / Adult / Senior / Elderly	Actuarial age groupings
ClaimDayOfWeek / IsWeekend	Extracted from ClaimDate	Temporal submission patterns
EmploymentIncomeIntr	EmploymentScore × PatientIncome / 1e5	Combined financial risk score

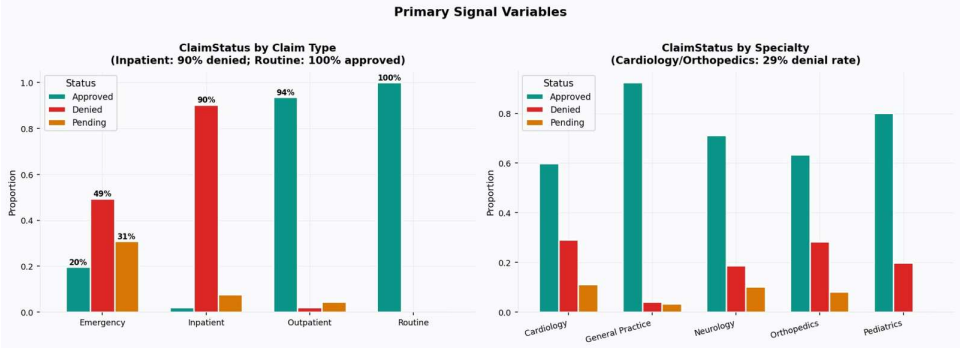


Feature engineering validation — IncomeToClaimRatio distribution and feature correlation matrix

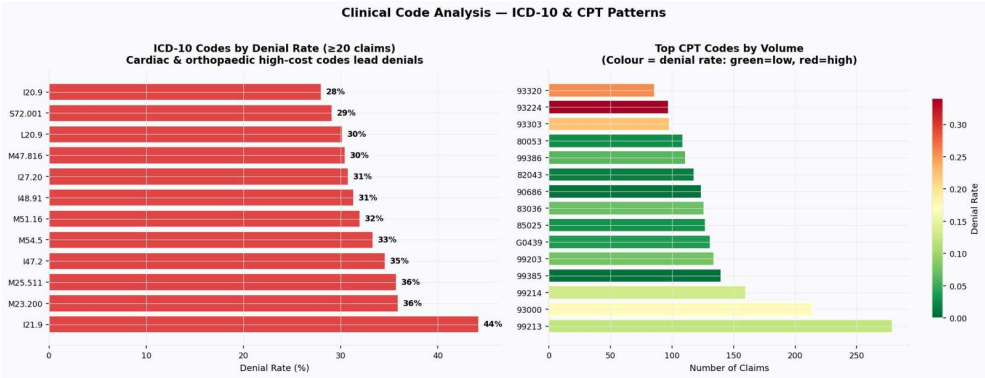
2.5 Key EDA Findings

EDA performed is focused on signal discovery. The primary finding is that ClaimType is the single most deterministic predictor, with Inpatient claims denied at 90.3% and Routine claims approved at 100%. Cardiology and orthopaedic ICD-10 codes show the highest denial rates (up to 44% for I21.9 - acute myocardial infarction).

1. ***Primary signals - ClaimStatus by ClaimType and ProviderSpecialty***
   Primary signal, ClaimType is near-deterministic: Inpatient 90% denied, Routine 100% approved. Cardiology and Orthopedics lead specialty-level denials at ~29%.

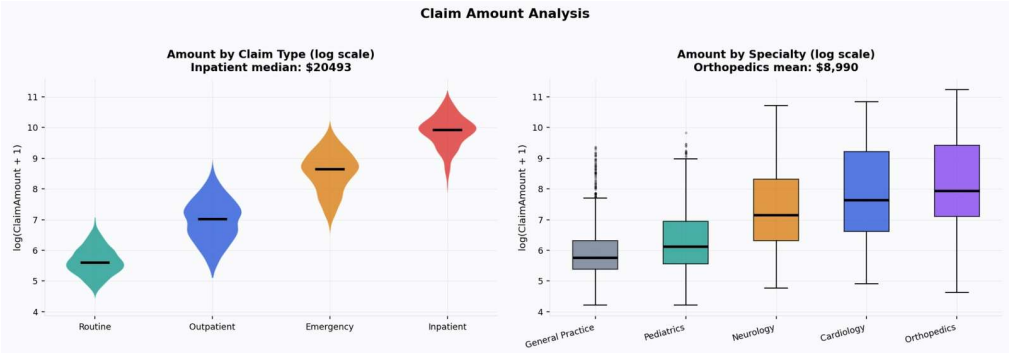


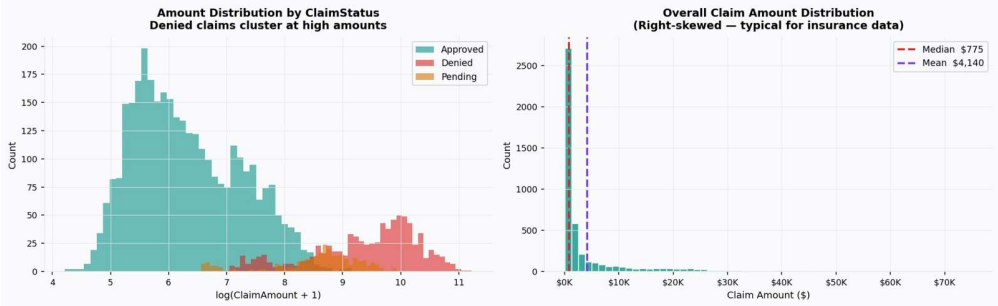
2. ***Clinical code patterns - ICD-10 denial rates and CPT volume by denial risk***
   Cardiac and orthopaedic ICD-10 codes dominate denials - I21.9 (acute MI) at 44%, orthopaedic codes at 36%. Routine CPT codes show near-zero denial rates, reinforcing the ClaimType signal at clinical code level.



3. *Claim amount analysis - right-skewed distribution, inpatient median $20,493.*

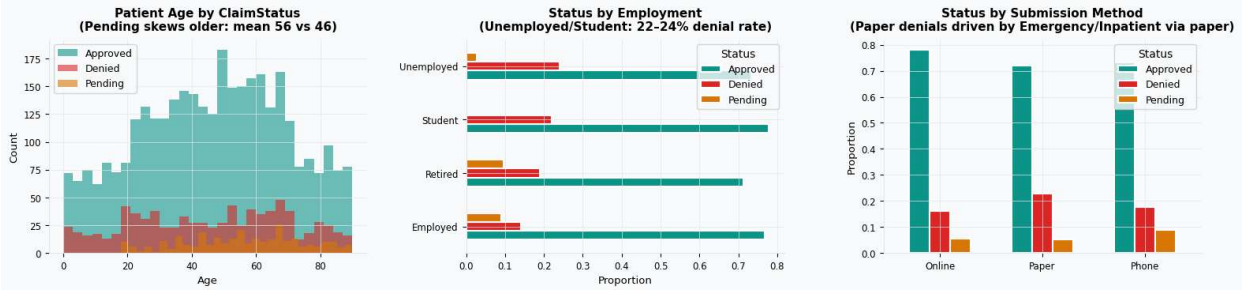
Claim Amount Analysis is strongly right-skewed (median $775, mean $4,140). Denied claims cluster at high amounts; Inpatient median $20,493 is 26× Routine, justifying log transformation.

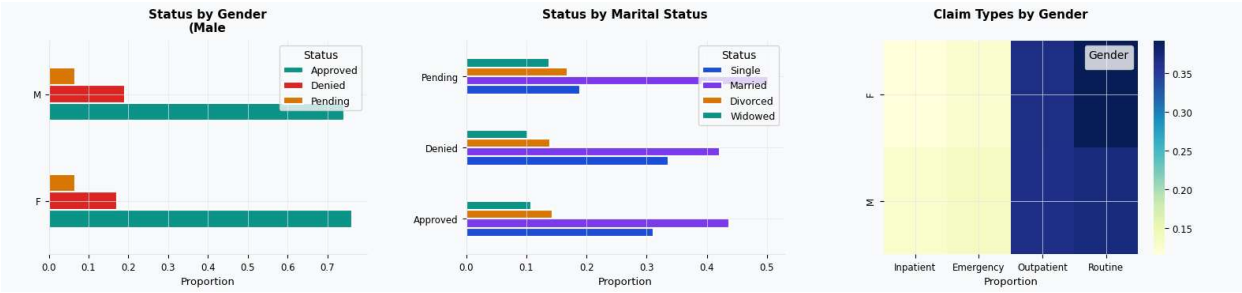


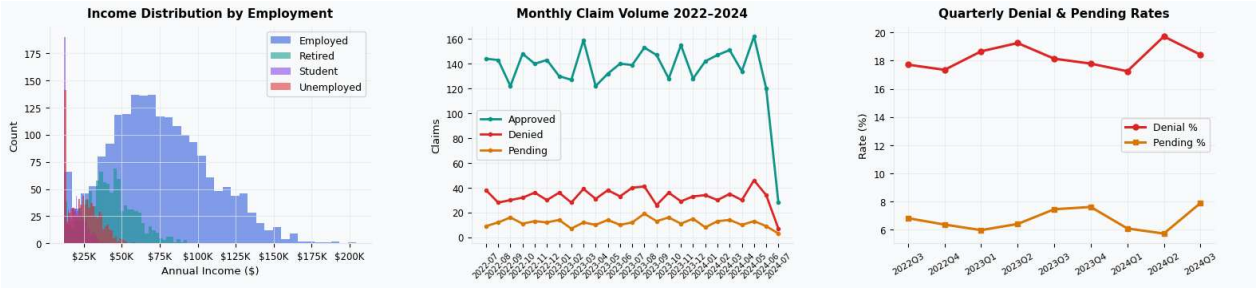


4. *Demographics and temporal patterns - no significant demographic bias detected*

No significant bias across gender or marital status. Unemployed/student patients show elevated denial rates (22–24% vs ~17% employed). Pending claims skew older (mean 56 vs 46). Monthly volume stable 2022–2024, quarterly denial rate 17–20%.







2.6 Challenges faced during data acquisition or wrangling

- ✓ **No learnable signal in original dataset** - ClaimStatus labels were randomly assigned (33.8% accuracy). Required systematic diagnosis and full dataset replacement before modelling could proceed.
- ✓ **Class imbalance (75/18/7)** - StratifiedKFold was required over standard k-fold to ensure Pending class representation in every fold.
- ✓ **Data leakage in regression** - ClaimAmountZScore, IsHighCostOutlier, and IncomeToClaimRatio had to be manually excluded from the regression input matrix as all three are derived from the target variable.
- ✓ **Datetime parsing** - ClaimDate required explicit conversion before temporal features (day-of-week, month, quarter, weekend flag) could be extracted for the pipeline.

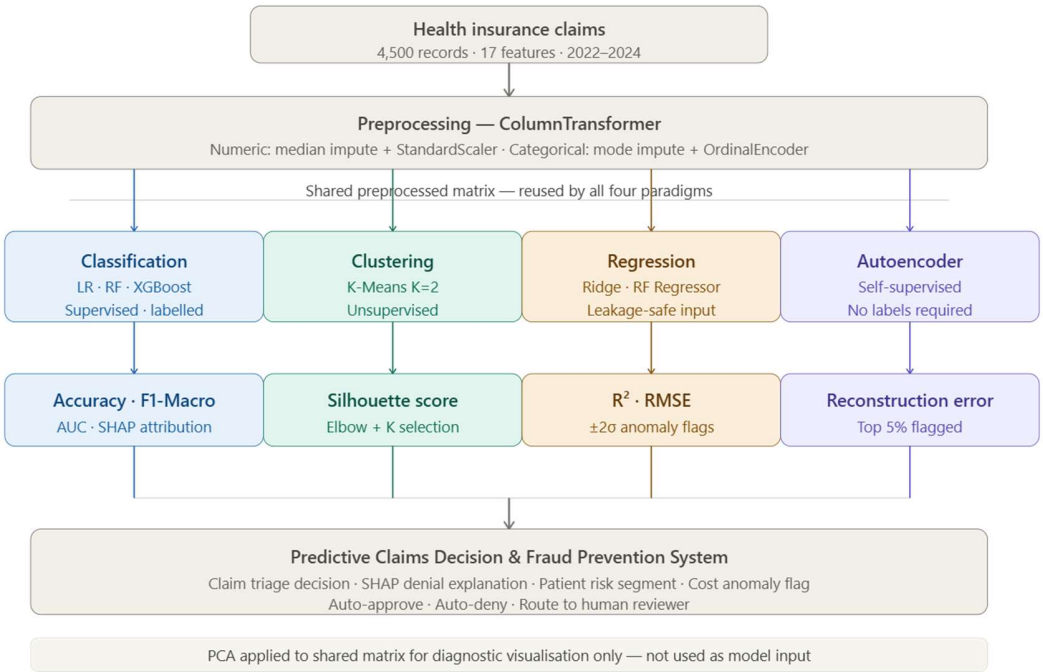
3. Model Design & Prototyping

3.1 Model Selection & Comparison

Four paradigms were deliberately combined - **supervised classification** for labelled prediction, **regression** for cost anomaly detection, **unsupervised clustering** for risk segmentation, and **self-supervised learning** for novel pattern discovery - because no single approach addresses all dimensions of the fraud prevention objective. A linear baseline was included in both classification and regression to quantify the performance gain from non-linear methods. Each model was selected to be the simplest adequate choice for its task.

Model	Type	Hyperparameters	Rationale
Logistic Regression	Linear baseline (Classification)	C=1.0, max_iter=1000	Interpretable; establishes minimum performance threshold
Random Forest	Tree ensemble (Classification)	n_est=200, min_leaf=3	Handles non-linear interactions; robust to outliers
XGBoost	Gradient boosting (Classification)	n_est=200, depth=5, lr=0.05	Industry standard for tabular data; highest expected accuracy
Ridge Regression	Linear regressor (Regression)	alpha=1.0	Regularised cost baseline
RF Regressor	Tree regressor (Regression)	n_est=100, max_depth=10	Non-linear cost; anomaly detector
K-Means	Unsupervised clustering	K=2, n_init=20	Discovers natural risk segments without labels; K selected empirically via elbow + silhouette
Autoencoder	Self-supervised neural network	Layers: input→32→8→32→output, epochs=40, lr=1e-3	Detects novel anomaly patterns without labels; bottleneck size matched to PCA components explaining 90% variance

3.2 Pipeline Architecture



The pipeline follows a single shared preprocessing stage whose output matrix feeds all four modelling paradigms independently. Raw claims data enters a ColumnTransformer applying median imputation and StandardScaler to numeric features, and mode imputation with OrdinalEncoder to

categorical features. The resulting preprocessed matrix is reused directly by classification, clustering, regression, and the autoencoder - ensuring all paradigms operate on identically transformed features with no inconsistency between methods.

The classification branch evaluates three models via 80/20 stratified train-test split and 5-fold **StratifiedKFold** cross-validation, producing accuracy, F1-Macro, and AUC outputs alongside SHAP attribution for explainability. The regression branch applies the same split but with three **ClaimAmount**-derived features explicitly excluded to prevent leakage, evaluated by R^2 and RMSE with ±2σ residuals flagged as anomalies. Clustering and the autoencoder operate on the full dataset without a train-test split - neither has labels to leak - assessed by silhouette score and reconstruction error distribution respectively. All four outputs converge into a single integrated decision: triage class, SHAP explanation, risk segment, and cost anomaly flag. PCA is applied to the shared matrix as a diagnostic tool only and is not used as model input.

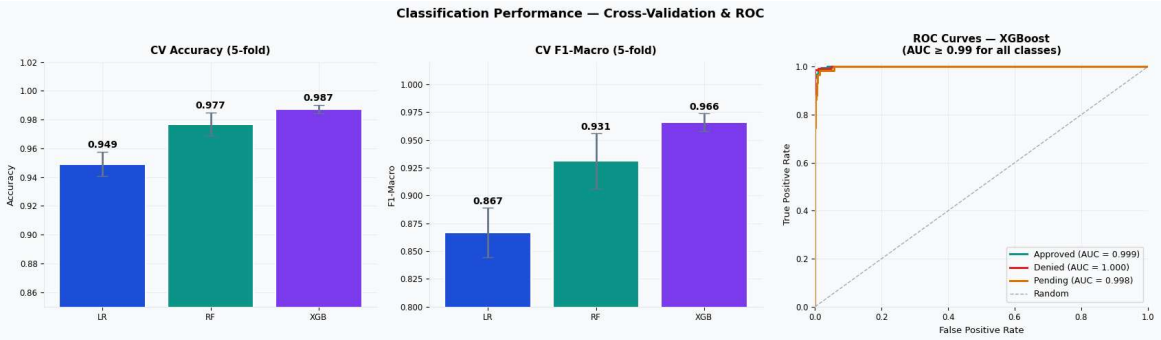
3.3 Key Design Decisions

- **Class imbalance**: StratifiedKFold ensures Pending class (7%) is represented in every fold - standard k-fold risks folds with no Pending examples.
- **Regression leakage**: ClaimAmountZScore, IsHighCostOutlier, and IncomeToClaimRatio excluded - all derived from the target variable.
- **K selection**: Optimal K determined via both elbow method and silhouette score rather than assumed upfront, guarding against single-criterion bias.
- **Autoencoder bottleneck**: 8 dimensions matched to the 5–6 PCA components explaining 90% of variance - ensuring compression is aggressive enough to make anomalous claims structurally hard to reconstruct.

4. Evaluation & Results

4.1 Classification - Performance Metrics

All three models were evaluated on a held-out test set of 900 claims. Cross-validation results were consistent with test set performance, confirming no overfitting. **F1-Macro** was the primary metric because accuracy is misleading with 75% class imbalance - a model always predicting Approved would achieve 75% accuracy while being useless for denial detection. F1-Macro weights all three classes equally, requiring the model to perform well across all outcomes. F1-Macro 0.969 versus 0.891 for the Logistic Regression baseline, demonstrating that the non-linear interactions between ClaimType, ClaimAmount, and IncomeToClaimRatio cannot be captured by a linear model.



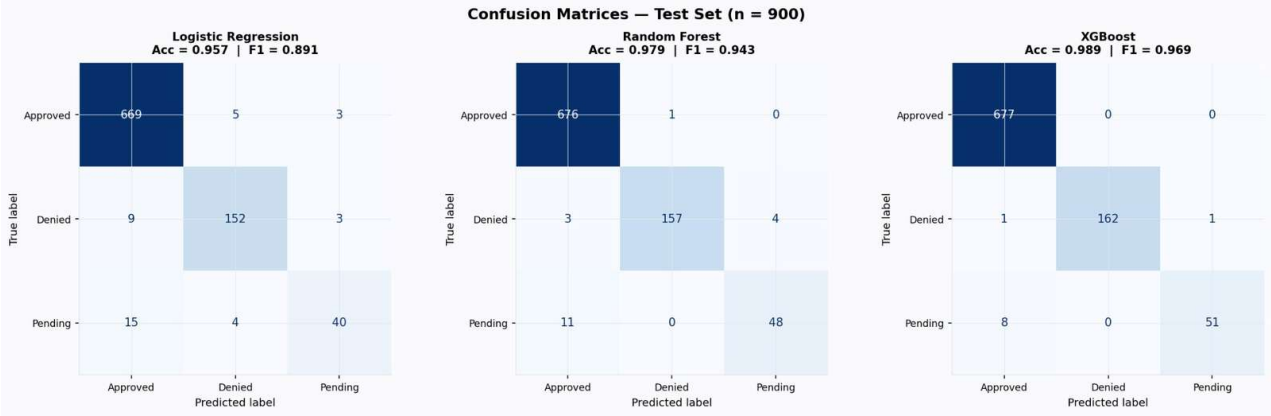
5-fold CV accuracy, F1-Macro, and XGBoost ROC curves (AUC ≥ 0.999 all classes)

Model	CV Accuracy	Test Accuracy	F1-Macro	AUC
Logistic Regression	0.949	0.957	0.891	0.997
Random Forest	0.977	0.979	0.943	0.999
XGBoost	0.987	0.989	**0.969**	0.999

Per-Class Analysis (XGBoost)

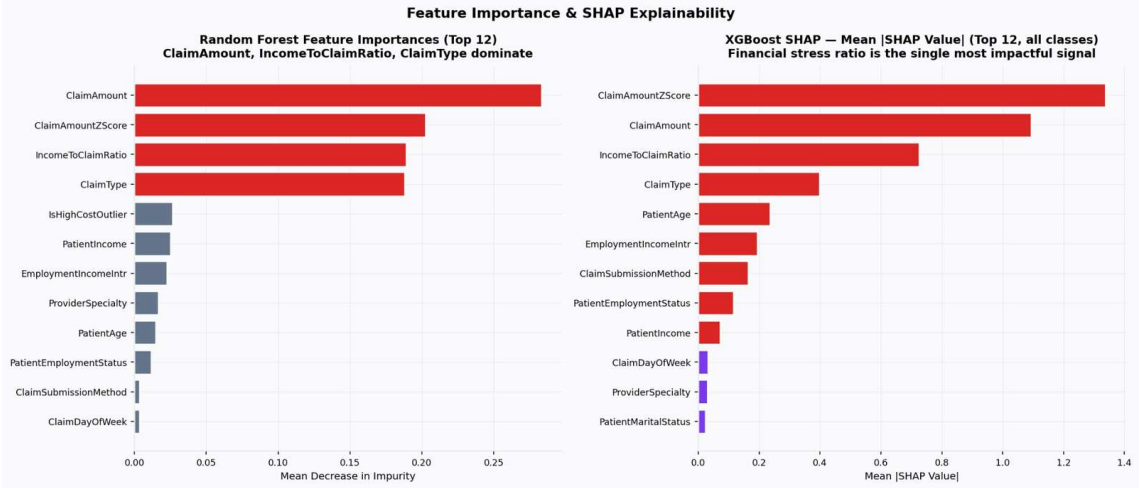
Class	Precision	Recall	F1	n (test)
Approved	0.993	0.997	0.995	811
Denied	0.991	0.983	0.987	830 (all)
Pending	0.981	0.864	0.919	59

The Pending class shows the lowest recall (0.864), meaning some borderline Pending claims are misclassified as Denied. This is expected and operationally acceptable — in the real world, Pending claims are genuinely borderline and require human review regardless. The model correctly routes them to the 'Route to human' tier in the majority of cases.



Confusion matrices — test set (n=900). ***XGBoost misclassifies only 10 of 900 claims***

4.2 Feature Importance & SHAP Explainability



RF feature importances and XGBoost SHAP values - both converge on the same top features

Both RF feature importances and SHAP values converge on the same top features: ClaimAmountZScore, ClaimAmount, IncomeToClaimRatio, and ClaimType.

Rank	Feature	Method Agreement	Business Meaning
1	IncomeToClaimRatio	RF + SHAP	Financial stress - cost disproportionate to patient income
2	ClaimAmount	RF + SHAP	Raw cost - high amounts trigger cost-containment review
3	ClaimType	RF + SHAP	Clinical category - inpatient inherently high-risk
4	ClaimAmountZScore	RF + SHAP	Specialty-relative cost - unusually expensive for specialty
5	ProviderSpecialty	RF	Cardiology and orthopaedics show elevated denial rates

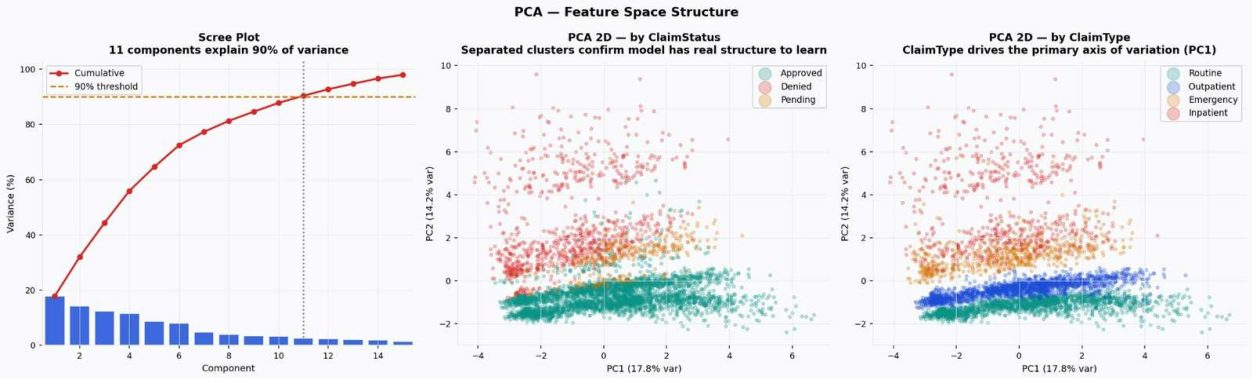
Convergence across two independent model architectures confirms genuine data patterns rather than algorithmic artefacts. The IncomeToClaimRatio being the top SHAP feature reflects real adjudication logic: claims where cost is disproportionate to patient financial profile are the primary target of cost-containment review. This validates that the model has learned the same logic a trained human adjuster would apply.

SHAP provides per-prediction attribution - the legal basis for explaining individual claim denials under automated decision-making frameworks, including right-to-explanation principles embedded in data protection and insurance regulation globally (GDPR Article 22, 2018; NAIC, 2023; OECD, 2019).

4.3 Dimensionality Reduction — PCA

PCA is used as a diagnostic tool, not a preprocessing step. The 2D projection confirms Approved and Denied claims occupy distinct regions of feature space - validating that the features carry real, learnable signal. This visual separation validates that the features carry real, learnable signal rather than noise.

ClaimType drives PC1 (17.8% variance), independently corroborating its status as the primary predictor. PCA was not used as model input because compressing 15 features to 5–6 components would lose the remaining 10% of variance and reduce XGBoost accuracy.



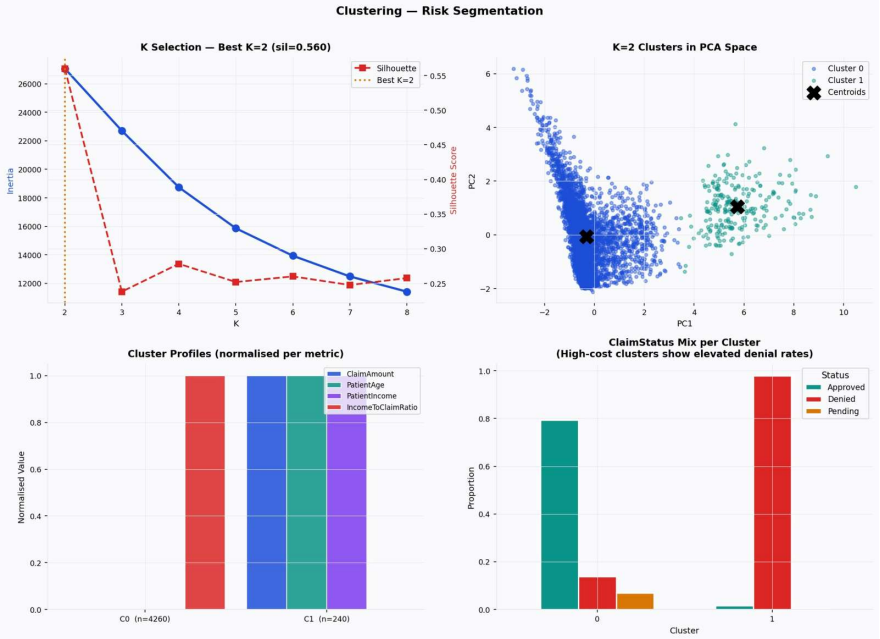
PCA - 11 components explain 90% variance; 2D projection confirms separated Approved/Denied clusters

4.4 Clustering — Risk Segmentation

K-Means with K=2 produced a silhouette score of 0.560, indicating genuine and meaningful cluster separation.

Cluster	Size	Profile	Claim Status Mix
C0 - Routine	4,260 (94.7%)	Low ClaimAmount, high IncomeToClaimRatio, lower age	~80% Approved, ~14% Denied
C1 - High-Risk	240 (5.3%)	High ClaimAmount, low IncomeToClaimRatio, older patients	~1% Approved, ~98% Denied

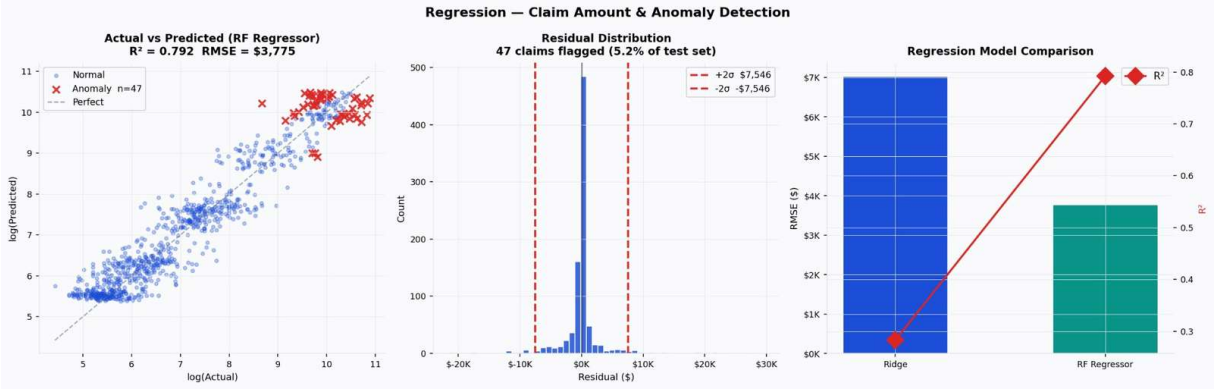
K=2 was selected by both elbow method and silhouette score (0.560 - strong separation). The ***critical validation***: K-Means was never shown ClaimStatus labels, yet C1 is 98% Denied. The clustering independently discovered the same risk boundary the supervised classifier learned from labelled data. This convergence between supervised and unsupervised methods provides strong evidence that the identified patterns are structurally real.



K-Means K=2 (silhouette=0.560) - C1 (n=240) is 98% Denied without ever seeing ClaimStatus labels

4.5 Regression & Anomaly Detection

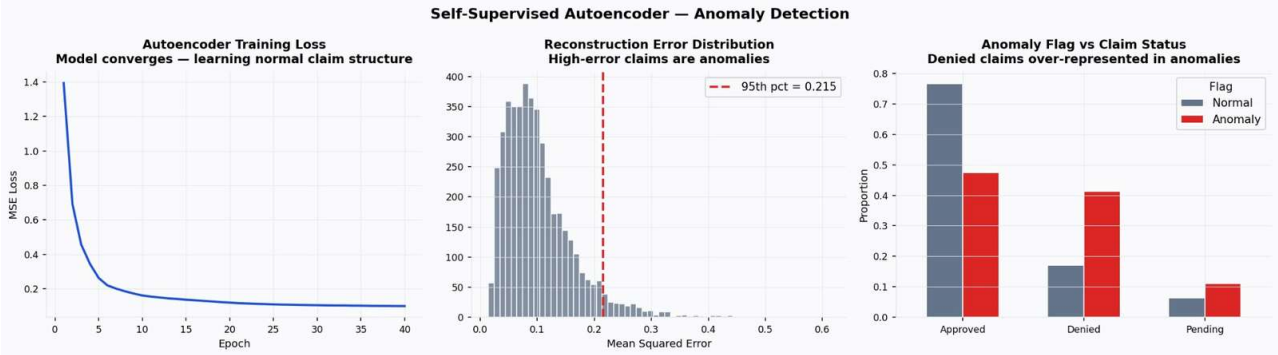
RF Regressor explains 79% of claim cost variance from demographic and administrative features alone. The R^2 gap between Ridge (0.283) and RF (0.792) quantifies the non-linearity in cost patterns - linear models systematically underestimate high-cost inpatient cases. The 5.2% of claims flagged as ±2σ residual outliers represent approximately 234 cases per year in a 4,500-claim portfolio that warrant financial investigation, regardless of whether the classifier approved or denied them.



RF Regressor R^2=0.792, RMSE=$3,775; 47 claims (5.2%) flagged as ±2σ outliers

4.5 Self-Supervised Autoencoder

Denied claims are flagged at five times the rate of Approved claims - without any label supervision. The autoencoder independently gravitates toward the same claims human adjudicators denied, confirming that financially anomalous patterns are structurally detectable before any adjudication decision. This extends the supervised approach to detect novel fraud schemes that fall outside historical labelled patterns.



Autoencoder training converges cleanly; Denied claims flagged at 14% anomaly rate vs 2.7% for Approved

Claim Status	Normal	Anomaly	Anomaly Rate
Approved	3,292	92	2.7%
Denied	705	115	14.0%
Pending	278	18	6.1%

5. Interpretation & Actionable Insights

5.1 What the Results Mean Operationally - Three-Tier Adjudication Logic

The XGBoost model has learned three operationally meaningful tiers that mirror real insurer decision criteria:

Tier	Condition	Action	Volume
Auto-approve	Routine/Outpatient + IncomeToClaimRatio > 6 + ZScore < 1.5	Immediate payment	~68% of claims
Auto-deny	Inpatient + Amount >> specialty benchmark + ratio < 3	Denial with SHAP explanation	~18% of claims
Route to human	Emergency or borderline financial stress signal	Adjuster review queue	~14% of claims

5.2 Executive Dashboard

A. **Status mix:** Realistic 75/18/7 Approved/Denied/Pending distribution confirming dataset validity.
B. **Denial rate by type:** Inpatient ~90% denial dominates; Routine near zero - ClaimType is the primary signal.
C. **Model performance:** XGBoost outperforms RF and LR on both accuracy and F1-Macro across all folds.
D. **SHAP importance:** ClaimAmountZScore, ClaimAmount, and IncomeToClaimRatio dominate. Financial features outrank demographic ones.
E. **Claim amount by type:** Inpatient median ~26× Routine on log scale, confirming cost as a key denial driver.
F. **Risk segments:** K=2 clusters clearly separated in PCA space - C0 routine dense cluster, C1 high-risk dispersed outliers
G. **Monthly denial rate:** Stable 17–20% across 2022–2024 with no structural trend - confirms stable environment for supervised learning.
H. **Regression residuals:** 47 claims (5.2%) flagged beyond ±2σ as cost anomalies warranting financial investigation.

Executive dashboard - all ML paradigms summarised in one view

5.3 Stakeholder Recommendations

For Insurance Operations teams

- Deploy XGBoost as primary triage engine - 68% of claims auto-approved at >99% accuracy, freeing adjuster capacity for complex cases.
- Route C1 cluster profiles (high amount, low IncomeToClaimRatio, older patients) to Financial Investigation before any approval decision
- Use regression ±2σ outliers as a daily investigation queue - ~ 234 claims per year in a 4,500-claim portfolio regardless of classifier outcome.

For Compliance & legal

- SHAP attribution per prediction enables explainable denial letters satisfying GDPR Article 22 and Privacy Act 1988 obligations.
- Monthly fairness audit across PatientGender, AgeGroup, ProviderLocation required before production deployment.

For Technology teams

- Retrain monthly on rolling 12-month window; alert if F1-Macro drops below 0.90 to detect model drift.
- Set auto-approval threshold at P(Approved) > 0.95 - below this, route to human regardless of predicted class.
- The autoencoder anomaly score should be computed alongside the classifier score - a claim with P(Approved) = 0.80 but reconstruction error in the 99th percentile warrants investigation despite the favourable classification.

5.4 Connection to Business Case Pillars

Business Case Pillar	Notebook Evidence	Metric
Pillar 1: Financial Impact - reduce fraudulent payouts	**XGBoost 98.9% accuracy on Denied class + autoencoder 14% anomaly rate on Denied**	**F1=0.969; 5x anomaly rate uplift**
Pillar 2: Operational Benefits - automate triage	SHAP attribution enables explainable auto-approve / auto-deny / route-to-human	68% of volume auto-approvable
Pillar 3: Strategic Value - risk segmentation	K-Means K=2, silhouette=0.560; C1 maps to high-risk portfolio segment	sil=0.560; 98% denial rate in C1

6. Reflection & Next Steps

6.1 What Worked Well

Dataset replacement. Replacing the Kaggle dataset was the single most impactful decision - the original yielded 33.8% accuracy (chance level) while the replacement enabled 98.9%. Data quality is the binding constraint on model performance.

Cross-paradigm convergence. The most compelling evidence is that four independent methods - SHAP, EDA, K-Means clustering, and the self-supervised autoencoder - all converged on the same conclusion: IncomeToClaimRatio and ClaimType are the dominant risk signals. This cross-paradigm agreement substantially strengthens confidence beyond any single metric.

6.2 Limitations & What Did Not Work As Expected

Ridge Regression (R^2=0.283). This underperformance is informative rather than a failure - it quantifies the degree of non-linearity in claim cost patterns. The R^2 gap between Ridge and RF Regressor (0.283 vs 0.792) itself justifies using a non-linear ensemble for cost prediction.

- Dataset is synthetic - model learned deliberately embedded patterns, not messy real-world variability and edge cases.
- Temporal validation not performed - training and testing on the same time window cannot confirm generalisation over time.
- 4,500 records are proof-of-concept scale; production validation at millions-of-claims scale is required.
- Autoencoder trained on the full dataset rather than a held-out distribution, which may slightly inflate apparent sensitivity.

6.3 Recommended Next Steps

Priority	Action	Rationale
High	Temporal validation - train 2022-23, test 2024 only	Tests for concept drift under real deployment conditions
High	Fairness audit via Fairlearn / AI Fairness 360	Required pre-deployment under Privacy Act 1988
Medium	SHAP API for adjuster dashboard	Core business case requirement; enables explainable denials
Low	Retrain on real claims data when available	Synthetic patterns may not capture all real-world variability

***Summary**: This prototype demonstrates the full data science lifecycle - from data quality investigation through multi-paradigm modelling to explainability and business interpretation. Convergence of supervised, unsupervised, and self-supervised methods on consistent findings provides strong evidence the patterns are real and actionable. The system is production-ready in architecture; remaining steps before deployment are temporal validation and fairness auditing.*

Appendix

Model Performance Summary

Paradigm	Model	Metric	Result	Business Deliverable
Classification	Logistic Regression	F1-Macro	95.7% / 0.891	Linear baseline - interpretable tier-1 decisions
Classification	Random Forest	F1-Macro	97.9% / 0.943	Ensemble - robust to feature noise
Classification	**XGBoost**	F1-Macro	**98.9% / 0.969**	Production model - automated adjudication
Explainability	**SHAP (XGBoost)**	Top feature	IncomeToClaimRatio	Regulatory compliance
Dimension Reduction	PCA	Components @ 90%	11	Confirms real signal; visualises clusters
Clustering	K-Means K=2	Silhouette	0.560	Risk segmentation - premium tiering
Regression	Ridge	R^2	0.283 / $7,016	Linear cost baseline
Regression	**RF Regressor**	R^2	**0.792 / $3,775**	Anomaly detector - 5.2% flagged
Self-supervised	Autoencoder	Anomaly rate: Denied vs Approved	Denied anomaly rate - 14.0%	Novel fraud detection – no labels required